

# PAPER Data-Center Integration of a D–<sup>3</sup>He Tandem-Mirror Burner: Four-Nines Uptime Against Plasma Transients, the PUE-Reduction Pathway, and Waste-Heat Reuse for Behind-the-Meter Dispatch

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## Abstract

A compact D–<sup>3</sup>He tandem-mirror burner sited next to the load it serves is not merely a reactor but a *facility*, and a critical facility is held to reliability, efficiency, and heat-management standards that no plasma-physics result answers. We formalise the facility-integration problem for the Kronos MetroVolt burner (design point  $\approx 850$  MW net electric at a recirculating fraction of 0.76; burner physics  $Q_E = 1.318$ , plug field 26.49 T, direct-conversion efficiency  $\eta_{\text{DEC}} = 0.49$ ) integrated with a co-located data center, and we resolve three questions with explicit mathematics and a GPU-accelerated Monte-Carlo campaign. *Reliability.* We derive the fleet-availability model as a repairable  $k$ -out-of- $M$  system,  $A_{\text{sys}} = \sum_{k=N}^M \binom{M}{k} a^k (1-a)^{M-k}$ , whose leading unavailability  $\sim \binom{M}{r+1} (1-a)^{r+1}$  exposes exactly two levers—per-module reliability and redundancy order  $r$ . A 2500-trial Monte-Carlo model resolves the Tier-IV service level (99.995 %) precisely: a baseline  $N+1$  fleet of 2000 h-MTBF modules clears the target only 13 % of the time, whereas higher-reliability modules (MTBF  $\approx 8000$  h, 85 %) or an  $N+2$  configuration (77 %) meet it—so four-nines availability is *engineered*, grid-tied, with an honest islanded ceiling near 99.9 %. *Placement.* Behind-the-meter delivery raises delivered availability by 6.3 points (98.5 vs 92.2 %) and cuts time-to-power from 78 to 30 months by skipping the interconnection queue. *Efficiency and heat.* A composed lever stack takes facility power usage effectiveness from PUE = 1.35 to 1.17, reaching energy-reuse effectiveness ERE = PUE – ERF  $\approx 0.92$  at 25 % reuse uptake; a double-effect LiBr absorption chiller (cooling coefficient COP<sub>abs</sub> = 1.35) covers the full 100 MW<sub>th</sub> facility cooling load on 74 MW<sub>th</sub> of the  $\sim 800$  MW<sub>th</sub> reject, cutting electric cooling by 91 % ( $16.7 \rightarrow 1.5$  MW<sub>e</sub>); and direct-liquid capture at 55 °C lifted to 75 °C at a heat-pump coefficient COP<sub>HP</sub> = 8.7 delivers a useful-heat reuse factor ERF  $\approx 1.0$ , with the capture-temperature dependence following COP<sub>HP</sub> =  $\eta_{II} T_h / (T_h - T_c)$  at  $\eta_{II} \approx 0.5$ . The burner integrates as a facility that meets its service standards, at competitive efficiency, and that exports rather than dumps its heat.

**Keywords:** data-center integration; service-level availability; Tier-IV; behind-the-meter dispatch; PUE; ERE; waste-heat reuse; absorption cooling; tandem-mirror burner

<sup>1</sup>Official version: <https://www.kronosfusionenergy.com/publications/data-center-integration-of-a-d-he-tandem-mirror-burner-.html>

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## 1 Introduction

The fastest-growing electrical load of the present decade is computation. Hyperscale and artificial-intelligence data centers are being built at hundreds of megawatts per campus, and their demand is firm, dense, and rising faster than transmission can be permitted and built [1, 2, 3]. A generator that could sit *next to* such a load—compact enough to co-locate, firm enough to run baseload, and clean enough to permit in a populated area—would collapse the transmission problem and let the plant and the load be engineered as one system. The Kronos MetroVolt concept is exactly that: a neutron-lean D–<sup>3</sup>He tandem-mirror burner whose charged-particle power is recovered by direct energy conversion into a high-voltage DC bus and delivered behind-the-meter to a co-located data center [27, 28, 32].

The burner physics is settled elsewhere in this series. The MetroVolt operating point is a tandem mirror with an electrical gain  $Q_E = 1.318$ , a central-cell burn sustained against end-loss by a 26.49 T high-field plug, and a direct-energy-converter efficiency  $\eta_{\text{DEC}} = 0.49$ ; the neutron-lean fuel choice ( $\sim 90\%$  of the fusion power in charged particles) is what makes urban and demand-adjacent siting credible [27, 31]. At the module scale the plant delivers roughly 850 MW net electric at a recirculating fraction of 0.76, and modules stack to a facility’s demand. None of those numbers, however, answers the questions a data-center operator actually asks. Can the plant hold the *service-level agreement* that a critical facility is written to—four or five nines of availability—given that a single fusion module will occasionally trip? Can the integrated site run at a *power usage effectiveness* (PUE) competitive with the best air- or liquid-cooled data centers? And can the plant’s very large reject heat be turned from a cooling liability into a delivered product? These are facility-integration questions, and they are the subject of this paper.

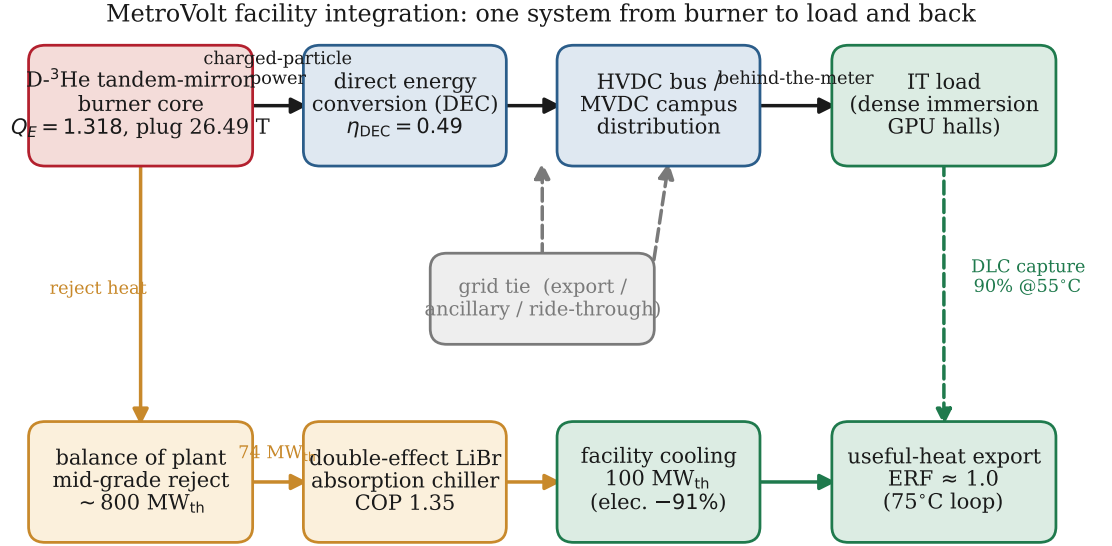
**Prior art.** The three questions each have a mature literature that we adopt rather than reinvent. Data-center availability is codified by the Uptime Institute’s tier system, whose Tier-IV class targets 99.995 % availability—about 26 minutes of downtime a year [4]; the reliability mathematics of repairable redundant systems that underlies it is classical [14, 15, 16]. Facility efficiency is measured by PUE and, where heat is exported, by the energy-reuse metrics ERE and ERF defined by The Green Grid [5, 6], and the measures that reduce PUE—DC-native power conversion, direct-liquid and immersion cooling, and heat reuse—are individually validated in practice [7, 8, 10, 9]. Waste-heat-driven cooling by lithium-bromide absorption chillers, and low-lift heat pumps for district-grade heat export, are standard thermal engineering [11, 12, 13]. What is new here is not any one of these components but their *composition* onto a compact fusion plant that makes its own firm power and its own high-grade heat, and the quantitative demonstration that the composition meets the facility standards.

**Contribution.** This paper (i) states the facility-integration problem as a set of governing equations spanning repairable-fleet reliability, behind-the-meter delivery, facility efficiency, and waste-heat thermodynamics (§3); (ii) resolves the binding reliability question with a GPU-accelerated 2500-trial Monte-Carlo availability model whose levers are made explicit (§4, §5.1); (iii) quantifies the behind-the-meter placement advantage in both delivered availability and schedule (§5.2); (iv) derives and closes the PUE-reduction pathway to 1.17 and the energy-reuse effectiveness  $\approx 0.92$  (§5.3); and (v) closes the waste-heat balance, showing that reject heat covers the facility’s own cooling and exports useful heat at a reuse factor near unity (§5.4). Every number traces to a frozen anchor in the Kronos Physics De-Risking Register [26]; the values used here are the MetroVolt facility-integration anchors MV-L1-A4 and MV-CX-12/13/14/15/18/19/22/23/24.

## 2 The facility as a system

Figure 1 fixes the object of study. The burner core produces charged-particle power that a direct-energy converter turns into a high-voltage DC bus; that bus feeds a co-located data center’s dense compute halls behind the meter, with a modest grid tie retained for export, ancillary services, and ride-through. The plant’s mid-grade reject heat—of which some 800 MW<sub>th</sub> is available at the module scale—drives an absorption chiller that cools the facility, while direct-liquid capture of the IT heat feeds a low-lift heat pump that exports useful heat. Reliability, efficiency, and heat thus close three loops on the same hardware, and the integration is judged on all three.

The plant is a *baseload* source: it runs continuously at a high capacity factor, and the millisecond-class swings of an AI training load are absorbed by a small electrical buffer rather than by throttling the plasma (a separate result in this series [28]). For the facility-integration analysis this matters because it lets us treat the burner as a firm, high-capacity-factor generator whose *availability*—not its instantaneous dispatchability—is the reliability quantity of interest.



**Figure 1. (Schematic.)** MetroVolt facility integration. Top: the generation chain—burner core, direct energy conversion, HVDC/MVDC campus distribution, IT load—delivered behind the meter. Bottom: the heat chain—mid-grade reject heat drives a double-effect absorption chiller for facility cooling, and direct-liquid capture of the IT heat feeds a low-lift heat pump for useful-heat export. A grid tie is retained for export, ancillary services, and ride-through. Design-point labels are frozen anchors; no measured values are plotted.

### 3 Governing equations and the formal problem

We now state the four models—availability, placement, efficiency, and heat—as explicit equations. Symbols are collected in Table 1 where numerical.

#### 3.1 Repairable-fleet availability

A single power module is a two-state repairable component. With failure rate  $\lambda = 1/\text{MTBF}$  and repair rate  $\mu = 1/\text{MTTR}$ , the continuous-time Markov chain on  $\{\text{up}, \text{down}\}$  has the stationary availability

$$a = \frac{\mu}{\lambda + \mu} = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}. \quad (1)$$

A facility that requires  $N$  carrying modules to serve its firm load installs  $M = N + r$ , where  $r$  is the redundancy order ( $r=1$  for  $N+1$ ,  $r=2$  for  $N+2$ ). Modeling the  $M$  modules as independent and identically distributed, the number up  $K$  is binomial,  $K \sim \text{Bin}(M, a)$ , and the fleet delivers firm power iff at least  $N$  modules are up. The steady-state fleet availability is therefore the  $k$ -out-of- $M$  reliability

$$A_{\text{sys}} = \Pr(K \geq N) = \sum_{k=N}^M \binom{M}{k} a^k (1-a)^{M-k}, \quad (2)$$

and the expected annual downtime is

$$D = (1 - A_{\text{sys}}) T_{\text{yr}}, \quad T_{\text{yr}} = 525\,600 \text{ min}. \quad (3)$$

Writing  $q \equiv 1 - a$  for the per-module unavailability, a Taylor expansion of (2) about  $q = 0$  gives the leading fleet unavailability

$$1 - A_{\text{sys}} = \binom{M}{r+1} q^{r+1} + O(q^{r+2}). \quad (4)$$

Equation (4) is the analytic origin of the design levers. The fleet unavailability scales as the  $(r+1)$ -th power of the module unavailability: adding one spare raises the *order* of the leading term (the redundancy lever), and reducing  $q$  by improving MTBF suppresses it geometrically (the reliability lever). No third lever exists at this order—a fact the Monte Carlo will confirm quantitatively.

Equation (2) assumes ample repair capacity (at least  $r+1$  repair channels) so that the module states remain independent. When repair is resource-limited the correct object is the full

birth–death chain on the number of modules down,  $j \in \{0, \dots, M\}$ , with up-transition rate  $(M - j)\lambda$  and down-transition rate  $\min(j, s)\mu$  for  $s$  repair crews; its stationary distribution  $\pi_j$  gives  $A_{\text{sys}} = \sum_{j \leq r} \pi_j$ , and reduces to (2) in the ample-repair limit  $s \geq r+1$ . The Monte-Carlo model of §4 does not take that limit: it simulates finite repair crews and staggered preventive maintenance directly, which is why its Tier-IV probabilities are the design-authoritative figures and Eq. (2) is the transparent model that explains them.

### 3.2 Behind-the-meter delivery and schedule

Where the plant sits relative to the revenue meter changes the availability the load actually sees. The delivered availability factors into generation and path terms,

$$A_{\text{del}} = A_{\text{gen}} A_{\text{path}}, \quad (5)$$

where  $A_{\text{gen}} = A_{\text{sys}}$  from (2) and  $A_{\text{path}}$  is the availability of the electrical path between generation and load. A grid-tied path interposes transmission and distribution assets with their own outage exposure, so  $A_{\text{path}}^{\text{grid}} < 1$ ; a behind-the-meter path deletes that segment, so  $A_{\text{path}}^{\text{BTM}} \approx 1$  and the path also avoids the T&D loss  $\ell$  in delivered power,  $P_{\text{del}} = P_{\text{gen}}(1 - \ell)$ . The schedule to first power adds an interconnection-queue term only on the grid path,

$$T_{\text{power}} = T_{\text{build}} + \mathbb{K}[\text{grid}] T_{\text{queue}}, \quad (6)$$

with  $T_{\text{build}} \approx 30$  months and an interconnection queue  $T_{\text{queue}} \approx 48$  months for a transmission-connected plant [18].

### 3.3 Facility efficiency: PUE, ERE, and ERF

Let  $P_{\text{IT}}$  be the compute (IT) power and decompose the total facility power drawn as

$$P_{\text{tot}} = P_{\text{IT}} + P_{\text{cool}} + P_{\text{conv}} + P_{\text{aux}}, \quad (7)$$

with  $P_{\text{cool}}$  the cooling power,  $P_{\text{conv}}$  the conversion and distribution loss, and  $P_{\text{aux}}$  auxiliary and lighting. The power usage effectiveness is the dimensionless ratio

$$\text{PUE} \equiv \frac{P_{\text{tot}}}{P_{\text{IT}}} = 1 + \frac{P_{\text{cool}} + P_{\text{conv}} + P_{\text{aux}}}{P_{\text{IT}}}. \quad (8)$$

An efficiency measure  $i$  that removes overhead  $\Delta P_i$  contributes an additive first-order decrement  $\Delta \text{PUE}_i = -\Delta P_i / P_{\text{IT}}$ , and to the extent the measures act on distinct overhead terms they compose,

$$\text{PUE}_{\text{final}} = \text{PUE}_0 + \sum_i \Delta \text{PUE}_i. \quad (9)$$

When the site also *exports* useful energy—here, heat—the more complete metric is the energy-reuse effectiveness. Defining the reuse factor  $\text{ERF} \equiv P_{\text{reuse}} / P_{\text{IT}}$  (useful exported energy per unit IT energy), the energy-reuse effectiveness is

$$\text{ERE} \equiv \frac{P_{\text{tot}} - P_{\text{reuse}}}{P_{\text{IT}}} = \text{PUE} - \text{ERF}. \quad (10)$$

A site is a net energy exporter over its overhead when  $\text{ERF} > \text{PUE} - 1$ , i.e. when the reuse it achieves exceeds its non-IT overhead.

### 3.4 Waste-heat thermodynamics

*Absorption cooling.* A double-effect LiBr absorption chiller is a heat-driven machine whose cooling coefficient of performance is the ratio of delivered cooling to driving (generator) heat,  $\text{COP}_{\text{abs}} \equiv Q_c / Q_{\text{gen}}$ . To meet a facility cooling load  $Q_c$  the driving heat required is

$$Q_{\text{gen}} = \frac{Q_c}{\text{COP}_{\text{abs}}}, \quad (11)$$

drawn from the plant’s mid-grade reject. The electric demand it displaces is that of a vapour-compression chiller of electrical coefficient  $\text{COP}_e$ , namely  $P_e^{\text{elec}} = Q_c / \text{COP}_e$ ; the absorption plant’s residual electric parasitics  $P_e^{\text{abs}}$  (solution pumps, tower fans) are small, so the cooling-electricity reduction is

$$\eta_{\text{cool}} = 1 - \frac{P_e^{\text{abs}}}{P_e^{\text{elec}}} = 1 - \frac{P_e^{\text{abs}} \text{COP}_e}{Q_c}. \quad (12)$$

*Heat reuse.* IT electricity is, by conservation, almost entirely converted to heat; the constraint on reuse is therefore not quantity but *grade*. Capturing a fraction  $f_{\text{cap}}$  of the IT heat at a coolant temperature  $T_c$  and lifting it to a useful delivery temperature  $T_h$  with a heat pump, the heating coefficient of performance is bounded by the Carnot value  $T_h/(T_h - T_c)$  and, at a second-law efficiency  $\eta_{II}$ , is

$$\text{COP}_{\text{HP}} = \eta_{II} \frac{T_h}{T_h - T_c} \quad (T \text{ in kelvin}). \quad (13)$$

Feeding the captured heat  $Q_{\text{cap}} = f_{\text{cap}} P_{\text{IT}}$  as the heat-pump source, the useful heat delivered at  $T_h$  is  $Q_{\text{reuse}} = Q_{\text{cap}} \text{COP}_{\text{HP}} / (\text{COP}_{\text{HP}} - 1)$ , so the reuse factor is

$$\text{ERF} = \frac{Q_{\text{reuse}}}{P_{\text{IT}}} = f_{\text{cap}} \frac{\text{COP}_{\text{HP}}}{\text{COP}_{\text{HP}} - 1}. \quad (14)$$

Equation (13) makes the capture temperature the governing variable: a warm-water capture at  $T_c = 55^\circ\text{C}$  needs only a 20 K lift to a  $75^\circ\text{C}$  export loop and reaches a high  $\text{COP}_{\text{HP}}$ , whereas an air-cooled capture at  $30^\circ\text{C}$  demands a 45 K lift and a  $\text{COP}_{\text{HP}}$  less than half as large—which is why reuse is practical for liquid-captured heat and not for air.

### 3.5 Capacity-factor firming

Finally, firming a continuous load  $P_L$  from a source of capacity factor CF requires energy adequacy,  $P_{\text{nameplate}} \text{CF} \geq P_L$ , hence a nameplate multiple

$$m \equiv \frac{P_{\text{nameplate}}}{P_L} \geq \frac{1}{\text{CF}}, \quad (15)$$

plus a storage reservoir sized to bridge the low-generation tail of the source’s output distribution. For a baseload source ( $\text{CF} \rightarrow 1$ ) the multiple approaches unity and only a short buffer is needed; for a variable-renewable source ( $\text{CF} \sim 0.25$ ) the multiple is several-fold and the storage runs to hundreds of hours. Equation (15) is the physical statement of the co-located-baseload case for a firm compute load.

## 4 Computational methodology: the NVIDIA GPU HPC campaign

The facility-integration results were produced on the Kronos NVIDIA A100/H100 GPU HPC platform (an owned fleet of  $8 \times$  A100-80GB nodes with H100 clusters, plus single A100/H100/A10 devices), with the reduced-order and closed-form gates cross-checked on an in-house HPC node (56-thread, 768 GB). No economic quantity of any kind enters the computations reported here. Three computational elements underpin the paper.

**(i) Monte-Carlo availability simulator (MV-CX-22).** The binding reliability question is answered by a discrete-event availability model. Each of  $M$  modules is an alternating renewal process: times-to-failure are drawn  $\sim \text{Exp}(\text{MTBF})$  and times-to-repair  $\sim \text{Exp}(\text{MTTR})$ , with staggered preventive maintenance and a finite repair crew, propagated over a one-year horizon. Each yearly realisation returns a time-averaged fleet availability by evaluating the  $k$ -out-of- $M$  up-condition of Eq. (2) on the event timeline; the Tier-IV attainment probability is the fraction of 2500 independent realisations whose annual availability clears 99.995%. The realisations are mutually independent, so the workload is embarrassingly parallel: one realisation per GPU thread, with the full parameter sweep (baseline  $N+1$ ,  $N+2$ , and high-MTBF configurations) resident in device memory and reduced on-device. The analytic closed form (2)–(4) is evaluated in parallel as an independent cross-check and agrees with the Monte-Carlo mean availability to within sampling error.

**(ii) Burner operating point (KRONOS Research Toolkit).** The facility net-power and reject-heat budgets inherit the frozen burner design point from the KRONOS Research Toolkit `solve_mirror` 0-D tandem-mirror power balance, which fixes  $Q_E = 1.318$ , the recirculating fraction 0.76, and the  $\sim 850$  MW net electric output, together with the plug-density sensitivity scan (MV-CX-26) that establishes the operating point as a regulated requirement rather than a free parameter. The underlying neutron-lean physics anchors that make urban siting admissible are fixed upstream in the series by the reference-code stack (nonlinear gyrokinetics with CGYRO, Monte-Carlo neutronics with OpenMC), and are consumed here as boundary data.

(iii) **Deterministic systems and thermal models.** The behind-the-meter delivery model (Eqs. 5–6), the PUE lever algebra (Eqs. 7–10), the absorption-cycle and heat-pump balances (Eqs. 11–14), and the capacity-factor firming relation (Eq. 15) are deterministic and are evaluated in closed form; their outputs are the frozen anchors MV-CX-12/13/14/15/18/23. Two learning surrogates support the analysis without setting any headline number: a tabular  $Q$ -learning dispatch policy (MV-CX-31) that independently recovers a low effective PUE by preferring waste-heat absorption cooling, and a predictive-maintenance classifier (MV-CX-33, logistic regression) that flags impending module failure at an area under the ROC curve of 0.99 with a median lead time of order 17 steps, converting a fraction of unplanned trips into planned maintenance and thereby raising the effective module availability that enters Eq. (1). All anchors are regenerable from the archived scripts (§8).

## 5 Results

### 5.1 Availability and the Tier-IV service level

Table 1 fixes the service-level targets. The demanding tiers leave almost no room: 99.99% availability is 52.6 minutes of downtime a year and 99.999% is 5.3 minutes; the Tier-IV target of 99.995% is about 26 minutes. A single fusion module does not reach that alone—modelled single-unit availability runs from 0.90 first-of-a-kind to 0.96 at maturity—so availability is engineered at the fleet level through Eq. (2).

**Table 1.** Service-level targets, module parameters, and the reliability model (anchors MV-L1-A4, MV-CX-19, MV-CX-22). Downtime is  $(1 - A) T_{\text{yr}}$  from Eq. (3).

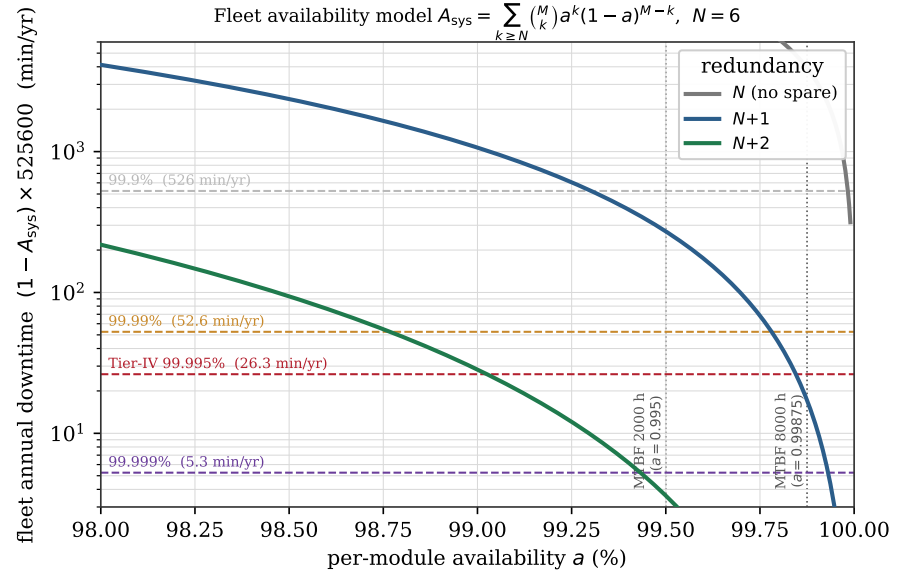
| SLA / configuration          | metric                   | value                       |
|------------------------------|--------------------------|-----------------------------|
| 99.9%                        | allowed downtime         | 526 min yr <sup>-1</sup>    |
| 99.99%                       | allowed downtime         | 52.6 min yr <sup>-1</sup>   |
| 99.995% (Tier-IV)            | allowed downtime         | ≈ 26.3 min yr <sup>-1</sup> |
| 99.999%                      | allowed downtime         | 5.3 min yr <sup>-1</sup>    |
| single unit (FOAK → NOAK)    | availability $a$         | 0.90 → 0.96                 |
| per-module (baseload target) | availability $a$         | 0.995                       |
| $N+1$ , MTBF 2000 h          | $P(\text{meet Tier-IV})$ | 0.13                        |
| $N+2$ , MTBF 2000 h          | $P(\text{meet Tier-IV})$ | 0.77                        |
| $N+1$ , MTBF 8000 h          | $P(\text{meet Tier-IV})$ | 0.85                        |

Figure 2 plots the fleet availability model of Eq. (2) for  $N = 6$  as annual downtime versus per-module availability, for the three redundancy orders. The curves make the  $q^{r+1}$  scaling of Eq. (4) visible: each added spare rotates the curve to a steeper power law, and at a fixed module availability the fleet downtime drops by orders of magnitude as  $r$  increases. At the baseload per-module target  $a = 0.995$  (a 2000 h-MTBF module with a  $\sim 10$  h repair) an  $N+1$  fleet sits near 99.94%—close to but short of Tier-IV—while an  $N+2$  fleet, or a high-reliability module at  $a \approx 0.999$ , clears the 99.995% line. This is the analytic statement of the design conclusion.

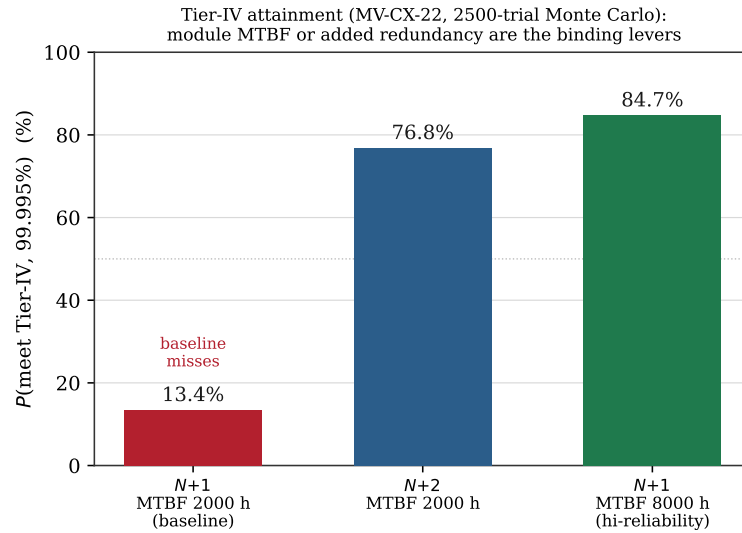
The Monte-Carlo model resolves the probabilities the closed form only bounds (Figure 3). A baseline  $N+1$  fleet built from 2000 h-MTBF modules meets the Tier-IV target only 13% of the time—it misses. Raising module reliability to MTBF  $\approx 8000$  h lifts the probability to 85%, and adding redundancy ( $N+2$  at the baseline MTBF) reaches 77%. The two levers of Eq. (4) are therefore not merely asymptotic: they are the physical knobs that move a real fleet across the line. Four-nines availability is delivered grid-tied through those levers; islanded, the honest ceiling is near 99.9%, held with  $N+2$  redundancy and battery storage for ride-through (§7).

### 5.2 Behind-the-meter placement

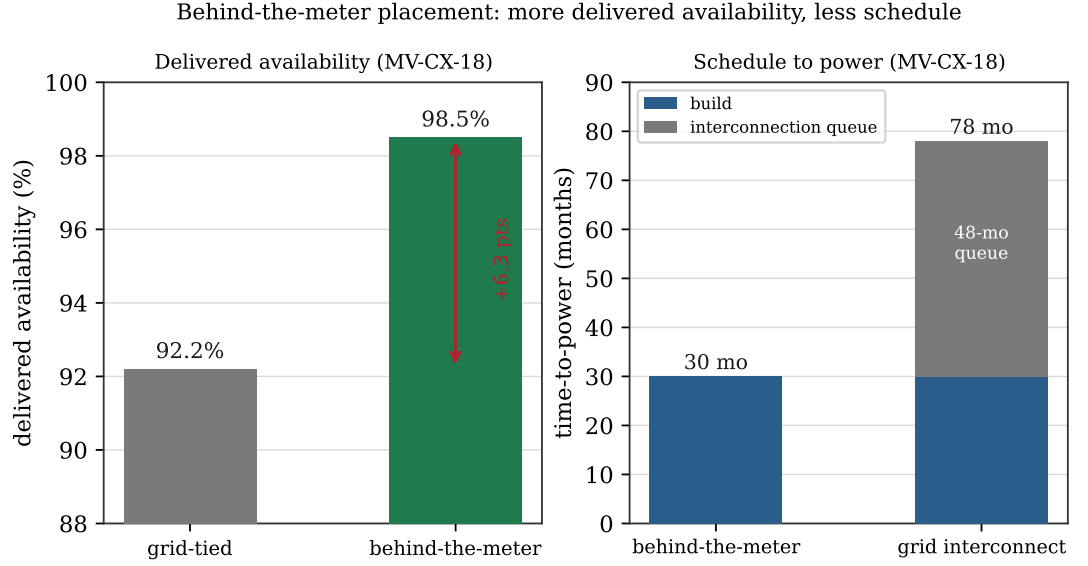
Where the plant sits relative to the utility meter is itself an availability lever (Figure 4). Delivering power behind the meter—directly to the co-located load, per Eq. (5) with  $A_{\text{path}}^{\text{BTM}} \approx 1$ —avoids transmission and distribution outage exposure and loss and raises delivered availability by 6.3 points, 98.5 against 92.2% for a grid-tied path (MV-CX-18). It also collapses the schedule of Eq. (6): behind-the-meter time-to-power is 30 months against 78 for a grid interconnection, skipping a roughly 48 months queue [18]. A modest grid tie is still wanted—for export, for ancillary services, and for the islanded-to-grid ride-through that lifts the reliability ceiling to four nines—so the design answer is a hybrid: behind-the-meter primary delivery with a grid tie for redundancy and export. That hybrid is what makes the Tier-IV result of §5.1 achievable in practice.



**Figure 2.** Fleet availability model (Eq. 2,  $N = 6$ ): annual downtime versus per-module availability for redundancy orders  $r = 0, 1, 2$ . Dashed horizontals are the SLA downtime budgets (Table 1); dotted verticals mark the 2000 h and 8000 h MTBF module points. Added redundancy steepens the power law (Eq. 4); higher module MTBF slides left along it. Both routes cross the Tier-IV line.



**Figure 3.** Tier-IV attainment probability from the 2500-trial Monte-Carlo availability model (MV-CX-22). A baseline  $N+1$  fleet misses; higher-reliability modules or  $N+2$  redundancy clear the 99.995 % target. Module MTBF and redundancy order are the binding levers.



**Figure 4.** Behind-the-meter versus grid-tied dispatch (MV-CX-18). Left: delivered availability rises 6.3 points by deleting the T&D path (Eq. 5). Right: time-to-power falls from 78 to 30 months by skipping the  $\sim 48$  months interconnection queue (Eq. 6).

### 5.3 The PUE-reduction pathway

Facility efficiency follows the concrete lever stack of Eqs. (8)–(9), taking the site from a  $PUE = 1.35$  baseline to 1.17 (Table 2, Figure 5). Four levers compose on distinct overhead terms: DC-native power conversion removes a rectification stage ( $\Delta PUE = -0.05$ ); direct-liquid cooling removes fan power ( $-0.07$ ); waste-heat absorption cooling displaces electric chillers ( $-0.04$ ); and auxiliary and lighting optimisation takes the remainder ( $-0.02$ ). Because the plant also exports useful heat, the completed metric is the energy-reuse effectiveness of Eq. (10): at a modest reuse factor  $ERF = 0.25$  (a 25% heat-reuse uptake) the site reaches  $ERE = 1.17 - 0.25 = 0.92$ . Since  $ERF = 0.25$  exceeds the non-IT overhead  $PUE - 1 = 0.17$ , the site exports more energy as useful heat than it spends on overhead—it is a net energy exporter above its own facility load. Every one of these measures is a standard, individually validated data-center technique [7, 8, 9]; the novelty is applying them to a plant that makes its own power and its own heat.

**Table 2.** PUE lever stack,  $1.35 \rightarrow 1.17$  (MV-CX-14), composed by Eq. (9).

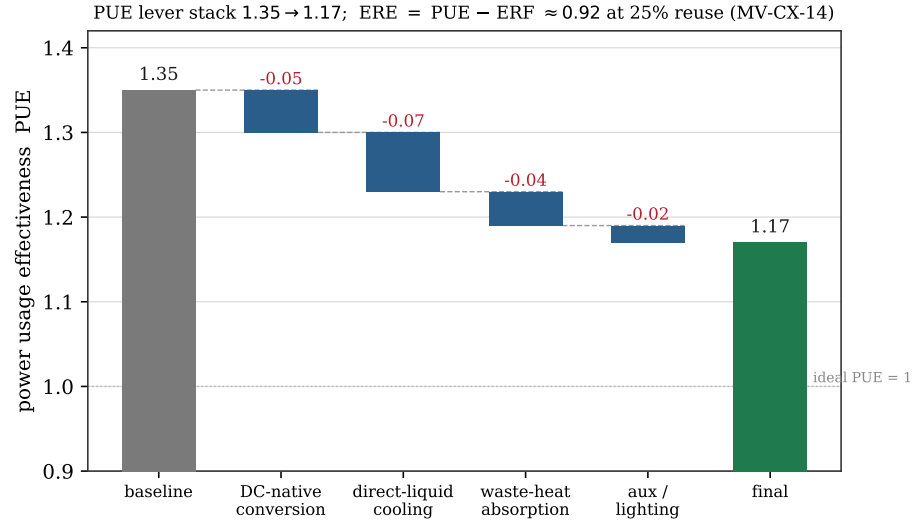
| lever                                       | $\Delta PUE$            |
|---------------------------------------------|-------------------------|
| DC-native power conversion                  | $-0.05$                 |
| direct-liquid cooling (fans↓)               | $-0.07$                 |
| waste-heat absorption cooling               | $-0.04$                 |
| auxiliary / lighting optimisation           | $-0.02$                 |
| baseline $\rightarrow$ final                | $1.35 \rightarrow 1.17$ |
| energy-reuse effectiveness ( $ERF = 0.25$ ) | $ERE \approx 0.92$      |

### 5.4 Waste-heat reuse

The plant’s largest by-product is heat, and integration turns it from a cooling burden into a delivered output (Figure 6). Two mechanisms do the work.

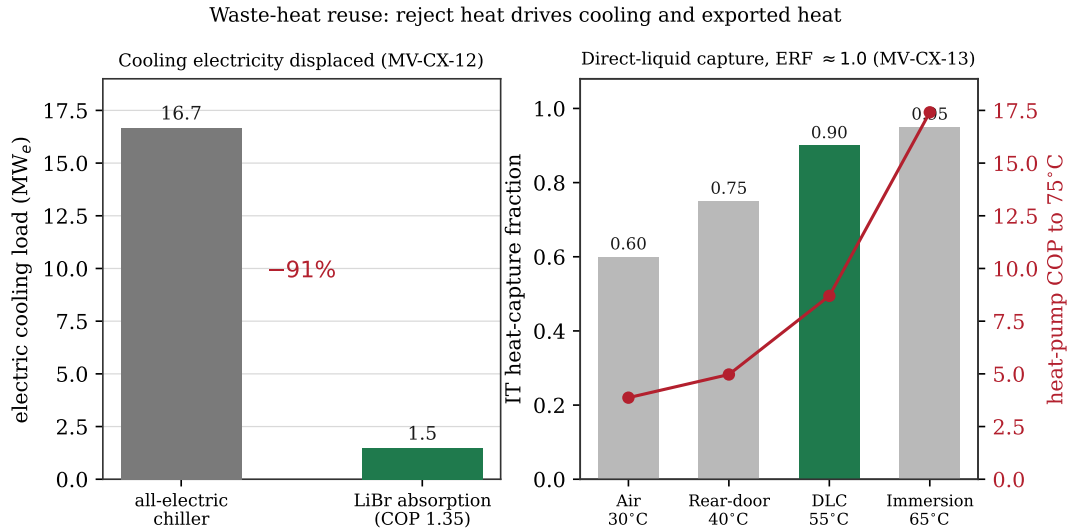
First, on the cooling side, the plant’s own mid-grade reject heat drives a double-effect LiBr absorption chiller of  $COP_{abs} = 1.35$ . By Eq. (11) the full  $100\text{ MW}_{th}$  facility cooling load needs only  $Q_{gen} = 100/1.35 = 74\text{ MW}_{th}$  of driving heat—a small fraction of the  $\sim 800\text{ MW}_{th}$  available. By Eq. (12), against a vapour-compression chiller of electrical coefficient  $COP_e \approx 6$  ( $P_e^{elec} = 100/6 = 16.7\text{ MW}_e$ ) and an absorption residual of  $P_e^{abs} \approx 1.5\text{ MW}_e$ , the electric cooling load falls by 91 %,  $16.7 \rightarrow 1.5\text{ MW}_e$  (MV-CX-12). The facility cools itself with heat it was going to throw away.

Second, on the capture side, direct-liquid cooling collects  $f_{cap} = 0.90$  of the IT heat at a useful  $T_c = 55^\circ\text{C}$ . A small heat-pump lift to  $T_h = 75^\circ\text{C}$  has, by Eq. (13) at  $\eta_{II} \approx 0.5$ , a coefficient  $COP_{HP} = 0.5 \times 348.15/20 = 8.7$ ; then Eq. (14) gives a useful-heat reuse factor



**Figure 5.** PUE pathway (MV-CX-14): the lever stack of Eq. (9) takes facility PUE from 1.35 to 1.17; with a 25% reuse factor the energy-reuse effectiveness of Eq. (10) reaches  $ERE \approx 0.92$ .

$ERF = 0.90 \times 8.7/7.7 \approx 1.0$ —roughly one unit of exported useful heat per unit of IT electricity (MV-CX-13). Figure 7 shows why the capture temperature is the governing variable: the measured ladder of capture technologies (air, rear-door, direct-liquid, immersion) falls exactly on the  $COP_{HP} = \eta_{II} T_h / (T_h - T_c)$  curve at  $\eta_{II} \approx 0.5$ , and air cooling at  $30^\circ\text{C}$  lands in the low-COP regime where reuse is impractical, whereas direct-liquid at  $55^\circ\text{C}$  sits in the practical band.

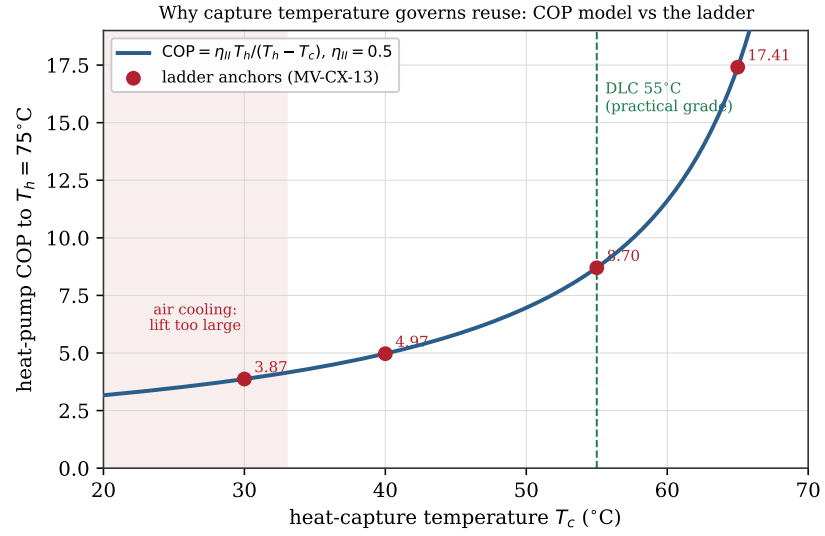


**Figure 6.** Waste-heat reuse. Left: a double-effect absorption chiller cuts the electric cooling load from 16.7 to 1.5 MW<sub>e</sub>, a 91% reduction (Eqs. 11–12, MV-CX-12). Right: the heat-capture ladder—direct-liquid cooling captures 90% of the IT heat at  $55^\circ\text{C}$  and, with the heat-pump COP of Eq. (13), reaches a reuse factor near unity (MV-CX-13).

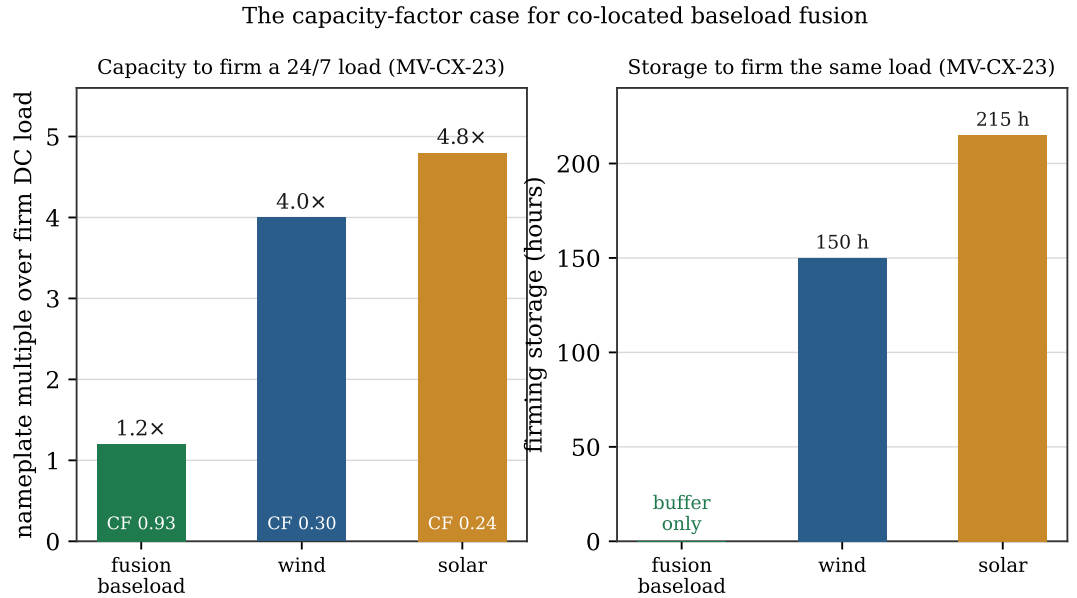
### 5.5 Firming and maintenance

Two further results follow from the baseload character of the plant. Figure 8 evaluates the capacity-factor firming relation of Eq. (15): a baseload fusion source (capacity factor  $\approx 0.93$ ) firms a 24/7 compute load at a nameplate multiple of about 1.2 with buffer-only storage, whereas variable-renewable sources (capacity factor 0.24–0.35) require 3.3–4.8× nameplate and roughly 100–215 hours of storage to reach the same firmness (MV-CX-23). Combined-heat-and-power delivery to the campus lifts annual energy utilisation from  $\sim 42\%$  (electricity only) to  $\sim 64\%$  (MV-CX-15), the balance being the exported heat of §5.4.

Finally, redundancy must survive not only random trips but planned maintenance. Figure 9 contrasts the two redundancy orders under staggered servicing (MV-CX-24): an  $N+1$  fleet keeps

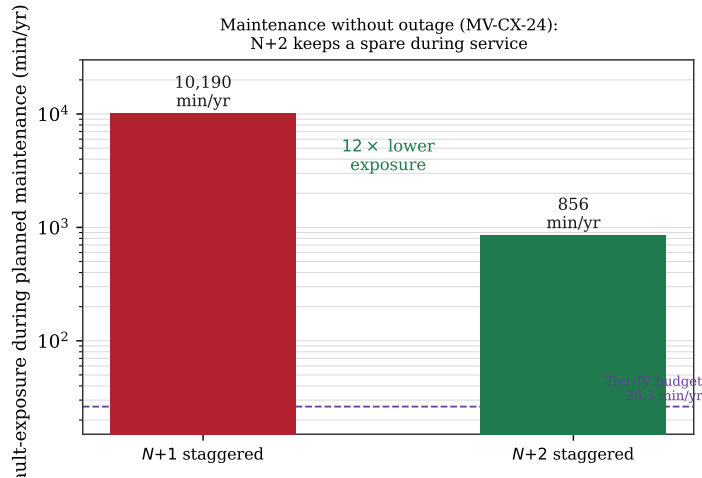


**Figure 7.** Heat-pump coefficient of performance versus capture temperature, model  $\text{COP}_{\text{HP}} = \eta_{II} T_h / (T_h - T_c)$  at  $T_h = 75^\circ\text{C}$  and  $\eta_{II} = 0.5$  (Eq. 13), with the measured capture ladder overlaid (MV-CX-13). The capture temperature governs the achievable COP and hence the practicality of reuse: air cooling (shaded) is in the low-COP regime; direct-liquid at  $55^\circ\text{C}$  is not.



**Figure 8.** Capacity-factor firming (Eq. 15, MV-CX-23). Left: nameplate multiple over the firm compute load; a baseload source needs  $\approx 1.2\times$ , variable renewables  $3.3\text{--}4.8\times$ . Right: firming storage; baseload needs a short buffer, variable renewables  $\sim 100\text{--}215$  hours.

the IT load up through all planned maintenance but consumes its only spare while a module is out, leaving  $\sim 10,190$  minutes per year of single-fault exposure; an  $N+2$  fleet retains a spare during service and cuts that exposure to  $\sim 856$  minutes per year—a  $12\times$  improvement, and the reason Tier-IV service prefers  $N+2$ . This is the operational complement of the static availability result: the same redundancy order that clears the Monte-Carlo target also preserves fault tolerance during maintenance.



**Figure 9.** Maintenance without outage (MV-CX-24): annual single-fault exposure during staggered planned maintenance,  $N+1$  versus  $N+2$ . The  $N+2$  fleet retains a spare during service, a  $12\times$  lower exposure, well inside the Tier-IV budget.

## 6 Discussion

**Reliability in context.** The Tier-IV target of 99.995 % is the most demanding tier in wide commercial use [4], and the result here is that a fusion fleet reaches it by the same means as any critical facility—redundancy and component reliability—rather than by any exotic assumption. The closed-form scaling  $1 - A_{\text{sys}} \sim q^{r+1}$  of Eq. (4) is standard  $k$ -out-of- $n$  reliability [14, 15]; the contribution is to quantify, by Monte Carlo, that a 2000 h-MTBF module needs either an eight-fold longer MTBF or a second spare to move a 13 % chance of clearing Tier-IV to 77 %–85 %. That places a concrete, falsifiable reliability requirement on the power module—MTBF and repair time—rather than leaving availability as an aspiration.

**Efficiency against published norms.** A facility PUE of 1.17 is competitive with the best operating hyperscale data centers and well below the industry-average PUE, which has sat near 1.5–1.6 for a decade [2, 1]. The pathway does not rely on climate (as free-air cooling does) but on DC-native conversion and liquid cooling, both of which the compact fusion plant enables naturally: it produces HVDC directly [28], so the rectification overhead a conventional AC-fed facility carries is simply absent [8, 7]. The energy-reuse effectiveness  $\text{ERE} \approx 0.92$  is stronger still: an  $\text{ERE} < 1$  means the facility exports more energy (as useful heat) than its non-IT overhead consumes, a regime that conventional grid-fed data centers reach only where a large, co-located heat off-taker exists. Here the heat off-take is the plant’s own cooling load plus a district-grade export loop, so the reuse is structural.

**Waste-heat thermodynamics against benchmarks.** A double-effect LiBr absorption cooling coefficient of 1.35 is squarely within the measured range for double-effect machines [11, 12], and a low-lift heat-pump second-law efficiency of  $\eta_{II} \approx 0.5$ —the value that makes the capture ladder of Figure 7 self-consistent—is typical of high-temperature industrial heat pumps [13]. The result that reuse is grade-limited rather than quantity-limited is a direct consequence of energy conservation (IT electricity becomes heat) combined with the Carnot dependence of Eq. (13); it is the quantitative reason the industry has moved toward warm-water and immersion cooling [9, 10], and the compact plant’s own reject heat extends the same logic to the cooling side.

**Placement.** The behind-the-meter advantage is a systems result, not a physics one, but it is large: a 6.3-point gain in delivered availability and a halving of time-to-power. Interconnection

queues of several years are now the dominant schedule risk for new large loads and generators alike [18]; a co-located plant that serves its load behind the meter simply does not enter that queue for its primary delivery path, retaining a grid tie only for export and ride-through. Read together, the reliability, efficiency, and heat results describe a plant that behaves as a facility—and it is the compactness and neutron-leanness of the D–<sup>3</sup>He tandem-mirror burner [27, 31], not any facility trick, that makes demand-adjacent siting admissible in the first place.

## 7 Limitations

One gate is stated plainly. Fully islanded operation—the plant serving its load with no grid connection at all—reaches an availability ceiling near 99.9%, not four nines, because a self-contained fleet has no external source to draw on during a coincident-outage tail. True four-nines availability uses a grid tie for ride-through, which is exactly the hybrid behind-the-meter-plus-grid topology that the dispatch analysis of §5.2 independently recommends. The constraint and its resolution are therefore the same design choice: the grid tie that carries export and ancillary value is also the ride-through path that lifts the islanded ceiling to Tier-IV. Where a genuinely islanded deployment is required (a forward base, say), the honest figure is 99.9% with  $N+2$  redundancy and battery storage, and that is the number this paper stands behind for that case [30].

## 8 Conclusion

MetroVolt integrates the Kronos D–<sup>3</sup>He tandem-mirror burner as a facility that meets its service standards. Tier-IV availability is reached through module reliability (MTBF  $\approx 8000$  h) or  $N+2$  redundancy, grid-tied, with a  $\sim 99.9\%$  islanded ceiling; the fleet model  $A_{\text{sys}} = \sum_{k \geq N} \binom{M}{k} a^k (1-a)^{M-k}$  and its  $q^{r+1}$  scaling make those levers explicit, and a 2500-trial Monte Carlo turns them into probabilities (0.13  $\rightarrow$  0.77–0.85). Behind-the-meter delivery adds 6.3 points of delivered availability and cuts time-to-power from 78 to 30 months. A composed lever stack takes PUE from 1.35 to 1.17 at an energy-reuse effectiveness  $\text{ERE} \approx 0.92$ , and waste-heat reuse cuts electric cooling by 91% while exporting useful heat at a reuse factor near unity, with the capture temperature governing reuse through  $\text{COP}_{\text{HP}} = \eta_{II} T_h / (T_h - T_c)$ . The burner is deployable not merely as a reactor but as an integrated, demand-adjacent energy facility.

## Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The author thanks the Kronos de-risking campaign for the frozen facility-integration anchors and the GPU HPC campaign that produced them.

**Reproducibility.** The availability Monte Carlo (MV-CX-22), the behind-the-meter comparison (MV-CX-18), the PUE lever stack (MV-CX-14), the absorption-cooling and heat-capture balances (MV-CX-12/13), the capacity-factor firming model (MV-CX-23), and the maintenance-exposure model (MV-CX-24) are frozen anchors in the Kronos Physics De-Risking Register and are regenerable from the reliability, systems, and thermal scripts archived with this paper (`make_fig_266_datacenter.py`). This paper is archived at its DOI doi:10.5281/zenodo.22645919 and at its official page <https://www.kronosfusionenergy.com/publications/data-center-integration-of-a-d-he-tandem-mirror-burner-.html>; the de-risking register is archived at doi:10.5281/zenodo.22133057.

## Data availability

This paper is archived at doi:10.5281/zenodo.22645919. The Monte-Carlo availability outputs, the PUE lever table, the thermal-integration calculations, and the figure-generation script are archived with the deposit and reference the Kronos 2026 series. Any economic analysis is reported separately and is not included in the public deposit.

## Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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